

RECEIVING REPORT

Receiver No.: 18912754

Document Date: 10/28/2025
Posting Date: 10/28/2025
Expiration Date:

P.O.#: 5300130352
Item #: 3
Batch#: ACE0185954

Insp. Lot: 000001376835
Plant: ACE
Plant Description: Clearfield-Solumina
Vendor Name: HEXCEL CORP
Vendor Addr: 1913 N KING ST
SEGUIN
TX 78155

Quality Inspection

Unloading Point:
Vendor No.: 90069901
Buyer Name: Keith Larsen
SQAR Code: i
Project ID: ASBUX

Test Lab:
WBS:

Inspection Lot/Type:

Rev No.: 01

Material: MSRR4040-VAFR

Received UOM: LB

Base UOM: LB

Base UOM Received Qty: 1340.920

Vendor Mtl. Number:

Mtl.Description: SATIN WEAVE CARBON FIBRE REINFORCED BISM

Mtl.Specification:

Alt. Material #:

Matl. Basic Data Text:

MSRR4040-VAFR:SATIN WEAVE CARBON FIBRE REINFORCED BISM

Bill of Lading: 92SPS057997

Delivery Note: 92SPS057997

Receipt Created by: P02073

Received Qty: 1340.920

Head text:

Receipt Delivered To: _____

WAREHOUSE INFORMATION



Plt: ACE



Part Number: MSRR4040-VAFR



Qty: 1340.92



Receiver No: 18912754

Storage Bin:



Order Number:

Warehouse: LIN1



Stock Type: Q



WBS Element: A-46Z01-NW.N46Z01000714

Inventory WBS: Y-ROLLS

Bin:

Storage Location: ACED

Item Text:

SATIN WEAVE CARBON FIBRE REINFORCED BISMALEIMIDE PREIMPREGN

ATTENTION: The revision level for this part is: 4

NG PROJECT: 46001.IN.INV.0RAW, PRIME CONTRACT:

Ship via: Best Way.

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P.O.#: 5300130352
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SQAR Code and Inspection Details:

Additional line-item provisions (AFQA-0001):
QSB18 – Chemical, Physical, and/or Mechanical Test Reports
QSB19 – Material Certifications
QMC3 – Remaining Shelf Life
QMC5 – Environmental Controls
QMC6 – Environmental Packaging
QSB24 - Certificate of Conformance
QQS12 - Rolls Royce SABRe

Traceability Control: *****LOT*****

DRAWINGS <(>&<)> SPECIFICATIONS

SPECIFICATION	REV	TITLE
MSRR4040	4	SATIN WEAVE CARBON FIBRE
REINFORCED BISMALEIMIDE PREPREG		
MSRR9992	13	APPROVAL, INSPECTION AND
TESTING PROCEDURES FOR PRE-IMPREGNATES		
MSRR9974	4	ENVIRONMENTAL TESTING OF
MATERIALS		
RPS670/22	5	BISMALEIMIDE RESIN
IMPREGNATED CARBON FIBRE PREIMPREGNATE MOULDING PARAMETERS		
MLC104	63	APPROVED CONSUMABLE,
NON-METALLIC AND PROPRIETARY MATERIALS		

ALL ROLLS-ROYCE SPECIFICATION REVISIONS SHALL BE MAINTAINED TO THE LATEST AVAILABLE THROUGH THE ROLLS-ROYCE SUPPLIER MANAGER WEBSITE.

Notes to Supplier:

- Direct warp. Warp side up. Blue poly must be touching the warp side. Yellow poly must be touching the weft side.
- Blue poly must be spooled on the outside of the spool
- Material must be manufactured by Hexcel Corporation (West Valley City, UT) or Hexcel Composites (Dagneux, France) per MLC104
- End Customer Rolls Royce
- Continual Improvement SABRe 4 Section 10.3

TACT STUDY:

- * Use Hexcel T6006 method of tack testing for lot acceptance testing.

PACKAGING, LABELING, AND SHIPPING INSTRUCTIONS:

SHIP ACCORDING TO MSRR9992 SECTION 12, SABRe ED.4 AND PER BEST COMMERCIAL PRACTICE TO ASSURE ACCEPTANCE BY CARRIER AND SAFE DELIVERY AT DESTINATION.

THE MATERIAL SHALL BE PACKED IN CONTAINERS THAT SHALL PREVENT DAMAGE AND CONTAMINATION DURING NORMAL HANDLING, TRANSIT AND STORAGE.

EACH REEL SHALL BE SEALED IN A MOISTURE PROOF BAG AND ADEQUATELY SUPPORTED AND PROTECTED TO PREVENT DAMAGE, DISTORTION AND CONTAMINATION DURING NORMAL HANDLING, TRANSIT AND STORAGE.

EACH CONTAINER SHALL BE MARKED WITH THE APPROPRIATE HAZARD WARNING LABEL.

	Qty.	Insp.
--	-------------	--------------

Insp. Stamp No.

By Inspector

Qty. Rej.

Rej. No.

Stocked By

Date Stkd.	
------------	--

Stk. Loc.



Packing list

Document 92SPS057997
Date 28-Oct-2025
Printed date 28-Oct-2025

Pallet	Box	Batch	Serial	Selling qty	Net weight	Gross weight
		92M0108617	92M0108617013	96.12lb	96.11lb	96.11lb
		92M0108617	92M0108617014	78.85lb	78.85lb	78.85lb
		92M0108617	92M0108617015	87.49lb	87.48lb	87.48lb
		92M0108617	92M0108617016	86.70lb	86.70lb	86.70lb
		92M0108617	92M0108617017	87.09lb	87.09lb	87.09lb
		92M0108617	92M0108617018	87.49lb	87.48lb	87.48lb
		92M0108617	92M0108617019	87.49lb	87.48lb	87.48lb
		92M0108617	92M0108617020	87.49lb	87.48lb	87.48lb
		92M0108617	92M0108617021	103.96lb	103.96lb	103.96lb

Total per Batch 92M0101738	ACE 0185950	79.25lb	18912750-
Total per Batch 92M0105068	ACE 0185951	1,772.85lb	18912751-
Total per Batch 92M0107768	ACE 0185953	87.88lb	18912752-
Total per Batch 92M0108616	ACE 0185954	1,340.92lb	18912754
Total per Batch 92M0108617	ACE 0185957	1,830.91lb	18912757
Total per Item 111545		5,111.81lb	5,111.64lb 5,111.64lb

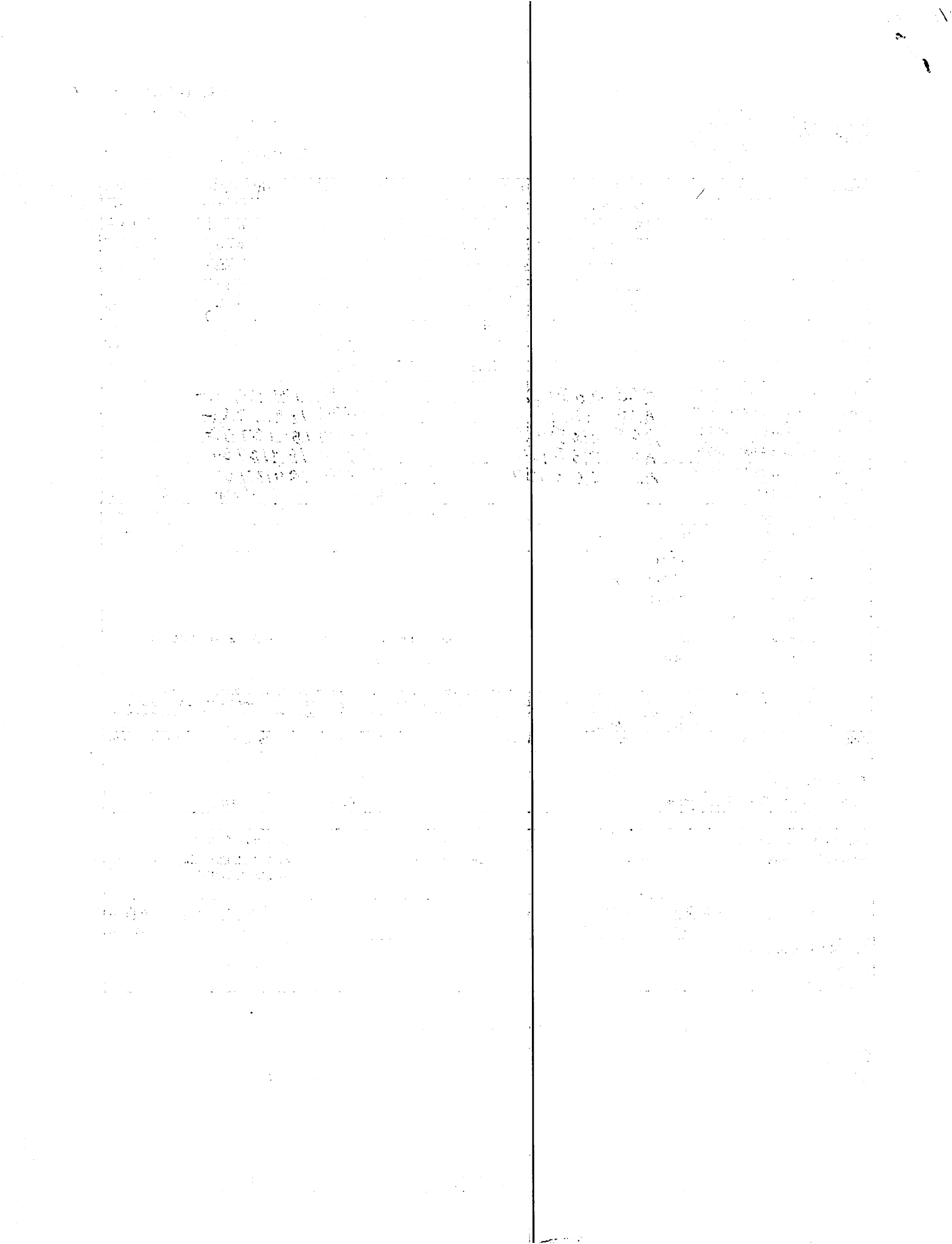
Line	Item	Description	Selling qty
6 3	92TempRec-Ambient	92TempRec-Ambient	2.00 ea
Customer part TEMREC			
Ship date 28-Oct-2025			
Sales line LotId 9216849146			
Trade compliance			
Country of origin RS		Commodity code 9025198085 / TEMPERATURE RECORDER	
ECCN EAR99		End user country US	
These items are controlled by the U.S. Government and authorized for export only to the country of ultimate destination for use by the ultimate consignee or end-user(s) herein identified. They may not be resold, transferred, or otherwise disposed of, to any other country or to any person other than the authorized ultimate consignee or end-user(s), either in their original form or after being incorporated into other items, without first obtaining approval from the U.S. government or as otherwise authorized by U.S. law and regulations.			

Pallet	Box	Batch	Serial	Selling qty	Net weight	Gross weight
				2.00ea	0.00lb	
Total per Batch				2.00ea		
Total per Item 92TempRec-Ambient				2.00ea	0.00lb	

Shipping Instructions				Carrier		Customer Pick-Up	
Mode of Delivery		Ground		Delivery Terms		FCA, Salt Lake City, UT, USA Incoterms®2010	
Pallet IDs	Box IDs	Batches	Serial Nos			Net weight	Gross weight
0	0	5	58			5,111.64lb	5,111.64lb
EU TMS Package details							
Line 5		0					
Line 6		0					

MSDS included with this shipment: 20396

Hexcel Corporation
6700 West 5400 South
West Valley City, UT 84118
United States





Packing list

Document	92SPS057997
Date	28-Oct-2025
Printed date	28-Oct-2025

MSDS included with this shipment: 20396

Hexcel Corporation
6700 West 5400 South
West Valley City, UT 84118
United States



Packing list

Document 92SPS057997
Date 28-Oct-2025
Printed date 28-Oct-2025

Customer P.O.



Customer P.O. 5300130352
Account C000079

Packing Slip ID



Hexcel order 92SO024797
Account C000079

Invoice address

Northrop Grumman Systems Corporation
Enterprise Transaction Services
8710 Freeport Parkway, Suite 200
Irving, TX 75063
United States

Delivery address

Northrop Grumman Systems Corporation
1051 S. Indust. Parkway
(ACCE)
Clearfield, UT 84015
United States

Delivery contact

Keith Larsen

Line	Item	Description	Selling qty			
5	111545	HexPly® F3C 433/M65;38%;285AW;50"	5,111.81-lb			
3						
Physical dimensions W = 50.00						
Customer part MSRR4040-VAFR						
Specification/Issue MSRR4040/Rev ISSUE 4						
Ship date 28-Oct-2025						
Sales line LotId 9216849145						
Test Results to Follow						
Trade compliance						
Country of origin US		Commodity code 6815130000 / Other articles of carbon fiber				
ECCN 1C010		End user country US				
These items are controlled by the U.S. Government and authorized for export only to the country of ultimate destination for use by the ultimate consignee or end-user(s) herein identified. They may not be resold, transferred, or otherwise disposed of, to any other country or to any person other than the authorized ultimate consignee or end-user(s), either in their original form or after being incorporated into other items, without first obtaining approval from the U.S. government or as otherwise authorized by U.S. law and regulations.						
Pallet	Box	Batch	Serial	Selling qty	Net weight	Gross weight
		92M0101738	92M0101738007	79.25lb	79.24lb	79.24lb
		92M0105068	92M0105068001	112.20lb	112.20lb	112.20lb
		92M0105068	92M0105068002	117.30lb	117.30lb	117.30lb
		92M0105068	92M0105068003	116.12lb	116.12lb	116.12lb
		92M0105068	92M0105068004	117.30lb	117.30lb	117.30lb
		92M0105068	92M0105068005	117.30lb	117.30lb	117.30lb
		92M0105068	92M0105068006	116.12lb	116.12lb	116.12lb
		92M0105068	92M0105068007	114.95lb	114.94lb	114.94lb
		92M0105068	92M0105068008	117.69lb	117.69lb	117.69lb
		92M0105068	92M0105068009	25.11lb	25.11lb	25.11lb

PGM7N02J60

PGN7N004C0

MSDS included with this shipment: 20396

Hexcel Corporation
6700 West 5400 South,
West Valley City, UT 84118
United States

1. The first part of the document is a list of names and addresses of the members of the committee. The names are listed in alphabetical order, and the addresses are listed in the order in which they appear in the list.

2. The second part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the chairperson of the committee.

3. The third part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the secretary of the committee.

4. The fourth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the treasurer of the committee.

5. The fifth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the clerk of the committee.

6. The sixth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant clerk of the committee.

7. The seventh part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant treasurer of the committee.

8. The eighth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant secretary of the committee.

9. The ninth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

10. The tenth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

11. The eleventh part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

12. The twelfth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

13. The thirteenth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

14. The fourteenth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

15. The fifteenth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

16. The sixteenth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

17. The seventeenth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

18. The eighteenth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

19. The nineteenth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.

20. The twentieth part of the document is a list of the names and addresses of the members of the committee who have been elected to the office of the assistant chairperson of the committee.



Packing list

Document 92SPS057997
Date 28-Oct-2025
Printed date 28-Oct-2025

Pallet	Box	Batch	Serial	Selling qty	Net weight	Gross weight
		92M0105068	92M0105068010	75.72lb	75.71lb	75.71lb
		92M0105068	92M0105068011	61.59lb	61.59lb	61.59lb
		92M0105068	92M0105068012	83.56lb	83.56lb	83.56lb
		92M0105068	92M0105068013	87.09lb	87.09lb	87.09lb
		92M0105068	92M0105068014	89.84lb	89.84lb	89.84lb
		92M0105068	92M0105068015	85.52lb	85.52lb	85.52lb
		92M0105068	92M0105068016	87.88lb	87.88lb	87.88lb
		92M0105068	92M0105068017	86.31lb	86.31lb	86.31lb
		92M0105068	92M0105068018	84.35lb	84.34lb	84.34lb
		92M0105068	92M0105068019	76.89lb	76.89lb	76.89lb
		92M0107768	92M0107768020	87.88lb	87.88lb	87.88lb
		92M0108616	92M0108616001	80.42lb	80.42lb	80.42lb
		92M0108616	92M0108616002	57.67lb	57.67lb	57.67lb
		92M0108616	92M0108616003	75.32lb	75.32lb	75.32lb
		92M0108616	92M0108616004	97.29lb	97.29lb	97.29lb
		92M0108616	92M0108616005	87.09lb	87.09lb	87.09lb
		92M0108616	92M0108616006	83.56lb	83.56lb	83.56lb
		92M0108616	92M0108616007	87.88lb	87.88lb	87.88lb
		92M0108616	92M0108616008	85.92lb	85.91lb	85.91lb
		92M0108616	92M0108616009	79.64lb	79.64lb	79.64lb
		92M0108616	92M0108616010	92.19lb	92.19lb	92.19lb
		92M0108616	92M0108616011	87.09lb	87.09lb	87.09lb
		92M0108616	92M0108616012	88.27lb	88.27lb	88.27lb
		92M0108616	92M0108616013	87.88lb	87.88lb	87.88lb
		92M0108616	92M0108616014	87.88lb	87.88lb	87.88lb
		92M0108616	92M0108616015	87.88lb	87.88lb	87.88lb
		92M0108616	92M0108616016	74.93lb	74.93lb	74.93lb
		92M0108617	92M0108617001	80.42lb	80.42lb	80.42lb
		92M0108617	92M0108617002	82.39lb	82.38lb	82.38lb
		92M0108617	92M0108617003	86.31lb	86.31lb	86.31lb
		92M0108617	92M0108617004	87.09lb	87.09lb	87.09lb
		92M0108617	92M0108617005	87.88lb	87.88lb	87.88lb
		92M0108617	92M0108617006	86.70lb	86.70lb	86.70lb
		92M0108617	92M0108617007	87.88lb	87.88lb	87.88lb
		92M0108617	92M0108617008	84.74lb	84.74lb	84.74lb
		92M0108617	92M0108617009	86.31lb	86.31lb	86.31lb
		92M0108617	92M0108617010	85.92lb	85.91lb	85.91lb
		92M0108617	92M0108617011	87.88lb	87.88lb	87.88lb
		92M0108617	92M0108617012	84.74lb	84.74lb	84.74lb

MSDS included with this shipment: 20396

Hexcel Corporation
6700 West 5400 South,
West Valley City, UT 84118
United States



28 October 2025

CERTIFICATE OF ANALYSIS

Customer: Northrop Grumman Systems Corporation

Purchase Order No: 5300130352 Part Number: MSRR4040-VAFR

Material: F3C 433/M65: 38%; 285AW; 50"

Specification: MSRR4040, Issue 4, dated November 2012.

Certificate of Conformance: Hexcel lot 92M0108616 has been manufactured and tested in conformance with specification MSRR4040, Issue 4, dated November 2012.

Lot No: 92M0108616 Date of Manufacture: 2 October 2025

Storage Life: 12 months from date of manufacture when stored at 0°.

Resin Lot No: 92M0108070 Manufactured by: Hexcel

Glass Lot No: 31W0028125 Manufactured by: Hexcel

- I. Prepreg Physical, Chemical and Mechanical Properties
See Certificate of Analysis Report.
- II. Individual Spool Physical Properties
See In Process Control Acceptance Testing Data.

Note: See Packing List for date of shipment, actual quantity and spools shipped.

Brandi Bond





Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0108616
Order Number: 92M0108616

Report Printed: 18-Oct-2025 12:57:43AM

Status: PASS
Approval: HOLTCH

1. Tensile, SBS, and TG Panel

Dimensional Fiber Volume

Roll/Panel	Test/Result:	Requirements:	Values:
92M0108616001/001	Laminate Density By Immersion Avg (g/cc)		1.5166
	)		
	Laminate Density By Immersion Std (g/cc)		0.0024
	Laminate Density By Immersion CoV		0.16
	Dimensional Fiber Volume (%)		53.14 - 53.95 - 53.10
	Dimensional Fiber Volume Avg (%)	50 - 56	53
	Dimensional Fiber Volume Std (%)		0.480
	Dimensional Fiber Volume CoV		0.90
	Void Volume (%)		0.4 - 0.5 - 0.6
	Void Volume Avg (%)	MAX= 2	0
	Void Volume Std (%)		0.100
	Void Volume CoV		19.90
92M0108616016/002	Laminate Density By Immersion Avg (g/cc)		1.5109
	)		
	Laminate Density By Immersion Std (g/cc)		0.0017
	Laminate Density By Immersion CoV		0.11
	Dimensional Fiber Volume (%)		53.47 - 53.20 - 52.64
	Dimensional Fiber Volume Avg (%)	50 - 56	53
	Dimensional Fiber Volume Std (%)		0.423
	Dimensional Fiber Volume CoV		0.80
	Void Volume (%)		0.9 - 0.8 - 0.8
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.040
	Void Volume CoV		4.77

Short Beam Shear Stress-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0108616001/001	SBS Strength (MPa)		84.1 - 81.1 - 84.8 - 78.5 - 83.2
	SBS Strength Avg (MPa)	MIN= 65	82
	SBS Strength Std (MPa)		2.56
	SBS Strength CoV		3.11
92M0108616016/002	SBS Strength (MPa)		76.5 - 76.2 - 73.2 - 79.4 - 80.4
	SBS Strength Avg (MPa)	MIN= 65	77
	SBS Strength Std (MPa)		2.85
	SBS Strength CoV		3.69

Notice: Values printed with '*' are out of Spec.

Page 1 of 4



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0108616**
Order Number: **92M0108616**

Report Printed: 18-Oct-2025 12:57:43AM

Status: **PASS**
Approval: **HOLTCH**

Short Beam Shear Stress-Warp-150C(302F)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0108616001/001	SBS Strength (MPa)		63.6 - 62.1 - 65.2 - 63.9 - 61.6
	SBS Strength Avg (MPa)	MIN= 50	63
	SBS Strength Std (MPa)		1.46
	SBS Strength CoV		2.31
92M0108616016/002	SBS Strength (MPa)		65.6 - 63.8 - 62.8 - 62.4 - 66.4
	SBS Strength Avg (MPa)	MIN= 50	64
	SBS Strength Std (MPa)		1.76
	SBS Strength CoV		2.75

Tg By DMA

Roll/Panel	Test/Result:	Requirements:	Values:
92M0108616001/001	Tg by DMA in °C (° C)		273.4
	Tg by DMA in °C Avg (° C)	MIN= 225.0	273.4
92M0108616016/002	Tg by DMA in °C (° C)		268.0
	Tg by DMA in °C Avg (° C)	MIN= 225.0	268.0

Calculated Thickness/Ply

Roll/Panel	Test/Result:	Requirements:	Values:
92M0108616001/001	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.29
	Calculated Thickness/Ply StDev (mm)		0.0051
	Calculated Thickness/Ply CoV		1.37
92M0108616016/002	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		0.91

Tensile Stress, Modulus, and Elongation-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0108616001/001	Norm Tensile Strength (MPa)		904.6 - 927.8 - 897.4 - 949.2 - 910.5
	Norm Tensile Strength Avg (MPa)	MIN= 700	918
	Norm Tensile Strength Std (MPa)		20.78
	Norm Tensile Strength CoV		2.26
	Norm Modulus (GPa)		60.9 - 61.9 - 61.8 - 61.8 - 61.7
	Norm Modulus Avg (GPa)	MIN= 55	62



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
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Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0108616
Order Number: 92M0108616

Report Printed: 18-Oct-2025 12:57:43AM

Status: PASS
Approval: HOLTCH

Tensile Stress, Modulus, and Elongation-Warp-Ambient (Continued)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0108616001/001	Norm Modulus Std (GPa)		0.379
	Norm Modulus CoV		0.62
92M0108616016/002	Norm Tensile Strength (MPa)		971.0 - 946.8 - 937.5 - 964.4 - 900.7
	Norm Tensile Strength Avg (MPa)	MIN= 700	944
	Norm Tensile Strength Std (MPa)		27.69
	Norm Tensile Strength CoV		2.93
	Norm Modulus (GPa)		61.7 - 61.9 - 62.7 - 62.6 - 62.1
	Norm Modulus Avg (GPa)	MIN= 55	62
	Norm Modulus Std (GPa)		0.414
	Norm Modulus CoV		0.67

2. PHYSICAL PROPERTIES

Volatile content @100°C in pan

Roll	Test/Result:	Requirements:	Values:
92M0108616001	Volatile content @100°C in a Pan (%)		0.5 - 0.4
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	0
92M0108616016	Volatile content @100°C in a Pan (%)		0.4 - 0.4
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	0

3. PHYSICAL TACK

Tack

Roll	Test/Result:	Requirements:	Values:
92M0108616001	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0108616003	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0108616005	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0108616007	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30

Notice: Values printed with "" are out of Spec.

Page 3 of 4



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0108616**
Order Number: **92M0108616**

Report Printed: 18-Oct-2025 12:57:43AM

Status: **PASS**
Approval: **HOLTCH**

Tack (Continued)

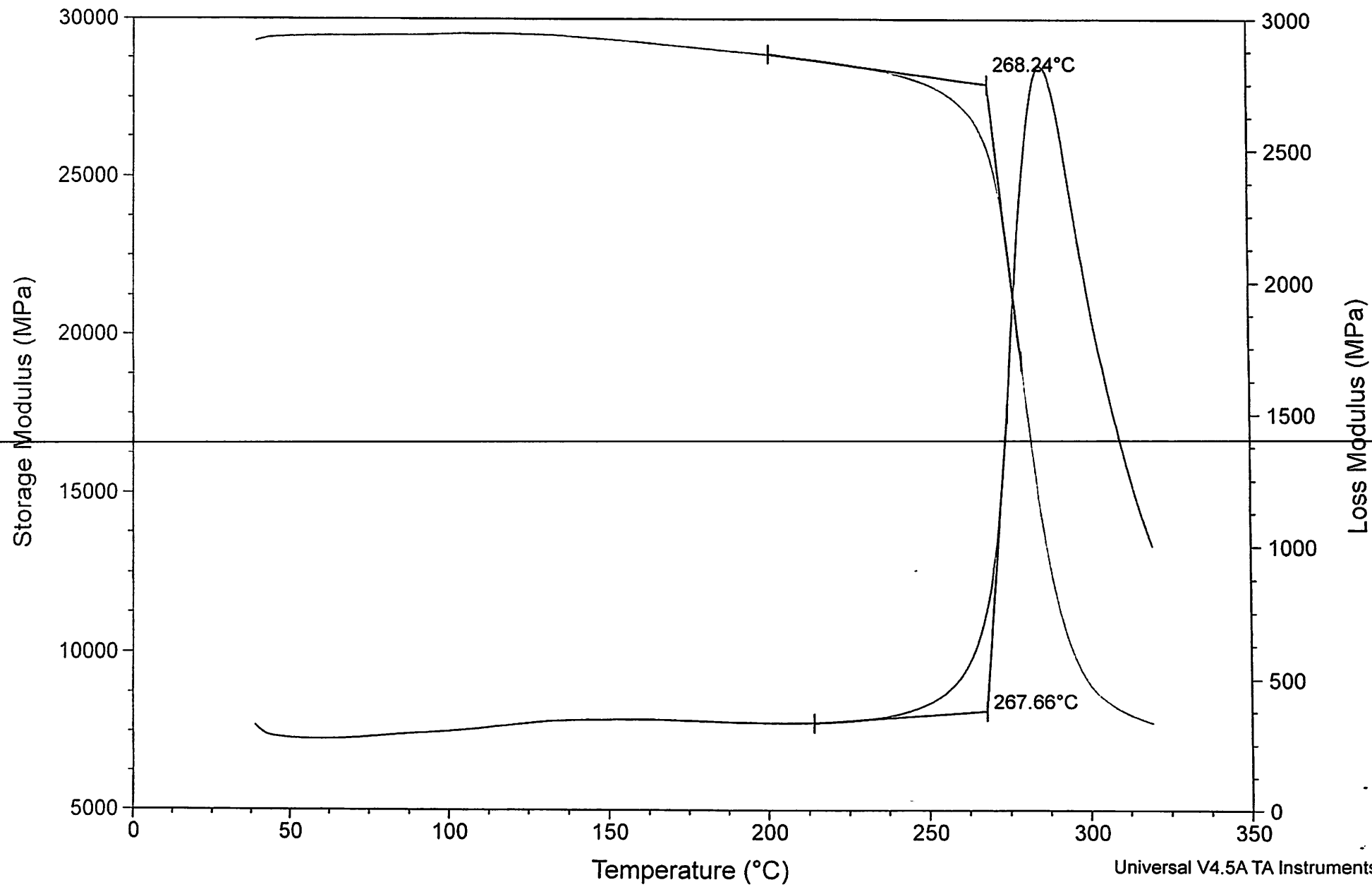
Roll	Test/Result:	Requirements:	Values:
92M0108616009	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0108616011	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0108616013	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0108616015	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30

End of report

Sample: 92M0108616-002 Sp016
Size: 34.9800 x 9.9200 x 2.3900 mm
Method: Temperature Ramp

DMA

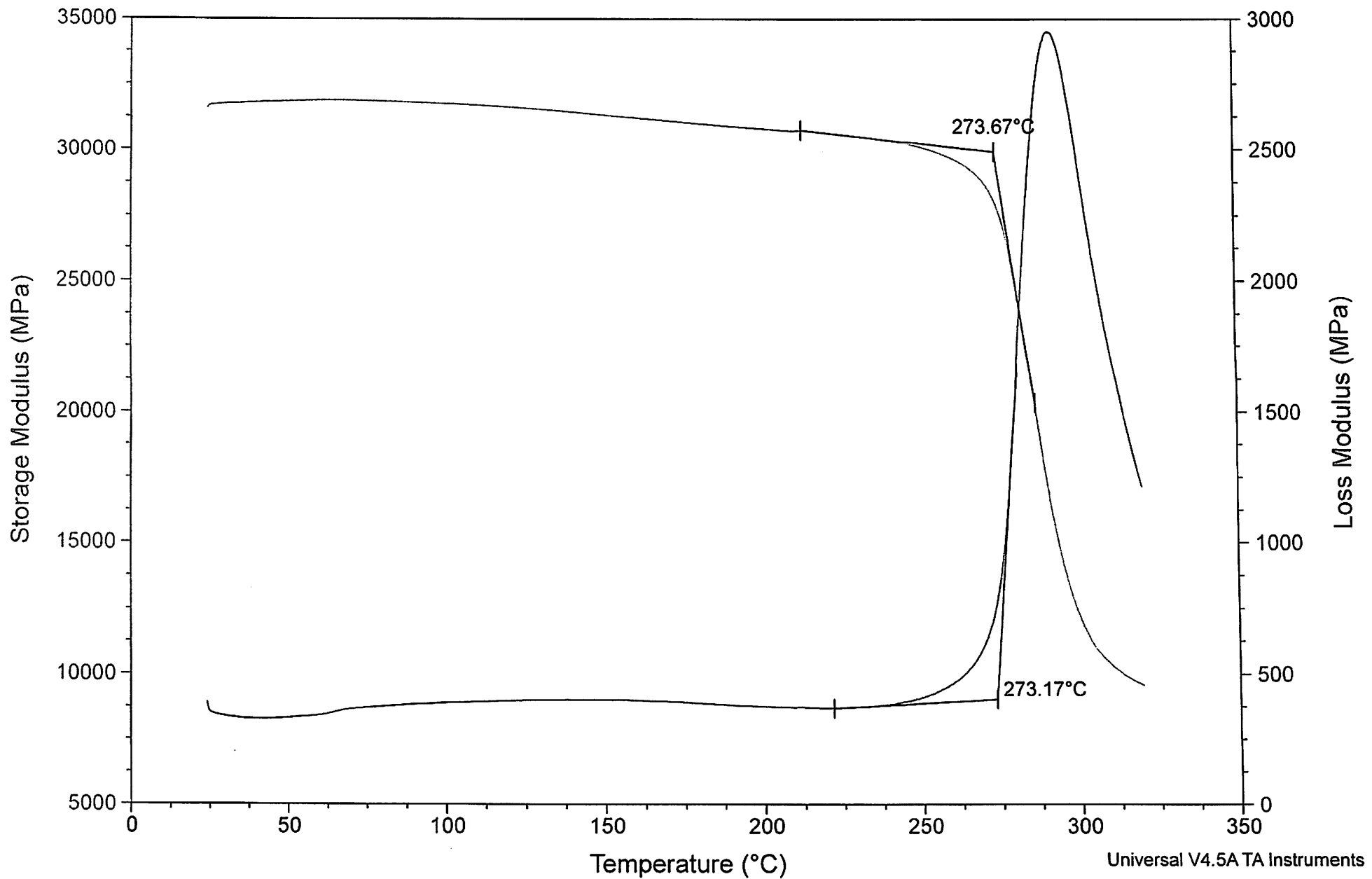
File: \\...\DMA Q800-2357\M56\92M0108616.002
Operator: RB
Run Date: 11-Oct-2025 22:00
Instrument: DMA Q800 V21.3 Build 96



Sample: 92M0108616-001 Sp001
Size: 34.9500 x 9.9100 x 2.2800 mm
Method: Temperature Ramp

DMA

File: \\...\\DMA Q800-2357\\M56\\92M0108616.001
Operator: RB
Run Date: 11-Oct-2025 20:30
Instrument: DMA Q800 V21.3 Build 96



IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: 92M0108616

10/27/2025

Page 1 of 3

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
92M 0108616001	0	1	4.5775	2.7957	38.9	280	457.8			TUIGAMPX	10/02/2025
		2	4.5577	2.7788	39.0	278	455.8			TUIGAMPX	10/02/2025
		3	4.5946	2.7917	39.2	279	459.5			TUIGAMPX	10/02/2025
		Avg.	4.5766		39	279	458	50.33	A	TUIGAMPX	10/02/2025
92M 0108616002	0	1	4.5712	2.8305	38.1	283	457.1			TUIGAMPX	10/02/2025
		2	4.5770	2.8394	38.0	284	457.7			TUIGAMPX	10/02/2025
		3	4.5611	2.8217	38.1	282	456.1			TUIGAMPX	10/02/2025
		Avg.	4.5698		38	283	457	50.33	A	TUIGAMPX	10/02/2025
92M 0108616003	0	1	4.6322	2.8583	38.3	286	463.2			TUIGAMPX	10/02/2025
		2	4.6615	2.8555	38.7	286	466.2			TUIGAMPX	10/02/2025
		3	4.6068	2.8389	38.4	284	460.7			TUIGAMPX	10/02/2025
		Avg.	4.6335		38	285	463	50.33	A	TUIGAMPX	10/02/2025
92M 0108616004	0	1	4.5930	2.8130	38.8	281	459.3			TUIGAMPX	10/02/2025
		2	4.5943	2.8255	38.5	283	459.4			TUIGAMPX	10/02/2025
		3	4.5724	2.8099	38.5	281	457.2			TUIGAMPX	10/02/2025
		Avg.	4.5866		39	282	459	50.33	A	TUIGAMPX	10/02/2025
92M 0108616005	0	1	4.6173	2.8254	38.8	282	461.7			TUIGAMPX	10/02/2025
		2	4.6060	2.8273	38.6	283	460.6			TUIGAMPX	10/02/2025
		3	4.6001	2.8190	38.7	282	460.0			TUIGAMPX	10/02/2025
		Avg.	4.6078		39	282	461	50.33	A	TUIGAMPX	10/02/2025
92M 0108616006	0	1	4.6191	2.8416	38.5	284	461.9			TUIGAMPX	10/02/2025
		2	4.5751	2.8439	37.8	284	457.5			TUIGAMPX	10/02/2025
		3	4.5908	2.8314	38.3	283	459.1			TUIGAMPX	10/02/2025
		Avg.	4.5950		38	284	460	50.33	A	TUIGAMPX	10/02/2025
92M 0108616007	0	1	4.6055	2.8064	39.1	281	460.6			TUIGAMPX	10/02/2025
		2	4.6162	2.8259	38.8	283	461.6			TUIGAMPX	10/02/2025
		3	4.6088	2.8212	38.8	282	460.9			TUIGAMPX	10/02/2025
		Avg.	4.6102		39	282	461	50.33	A	TUIGAMPX	10/02/2025
92M 0108616008	0	1	4.5790	2.8155	38.5	282	457.9			TUIGAMPX	10/02/2025
		2	4.5913	2.8219	38.5	282	459.1			TUIGAMPX	10/02/2025
		3	4.5631	2.8220	38.2	282	456.3			TUIGAMPX	10/02/2025
		Avg.	4.5778		38	282	458	50.33	A	TUIGAMPX	10/02/2025
92M 0108616009	0	1	4.5829	2.8147	38.6	282	458.3			TUIGAMPX	10/02/2025
		2	4.6312	2.8271	39.0	283	463.1			TUIGAMPX	10/02/2025
		3	4.5252	2.8180	37.7	282	452.5			TUIGAMPX	10/02/2025
		Avg.	4.5798		38	282	458	50.33	A	TUIGAMPX	10/02/2025
92M 0108616010	0	1	4.5879	2.8190	38.6	282	458.8			TUIGAMPX	10/02/2025
		2	4.5956	2.8380	38.2	284	459.6			TUIGAMPX	10/02/2025
		3	4.5517	2.8379	37.7	284	455.2			TUIGAMPX	10/02/2025
		Avg.	4.5784		38	283	458	50.33	A	TUIGAMPX	10/02/2025
92M 0108616011	0	1	4.5859	2.8185	38.5	282	458.6			TUIGAMPX	10/02/2025
		2	4.6138	2.8158	39.0	282	461.4			TUIGAMPX	10/02/2025
		3	4.5289	2.8100	38.0	281	452.9			TUIGAMPX	10/02/2025

ACCEPTANCE TESTING DATA

Page 2 of 3

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
=====											
			Avg. 4.5762		38	282	458	50.33	A	TUIGAMPX	10/02/2025
=====											
92M 0108616012	0	1	4.5954	2.8228	38.6	282	459.5			TUIGAMPX	10/02/2025
		2	4.6438	2.8527	38.6	285	464.4			TUIGAMPX	10/02/2025
		3	4.5666	2.8275	38.1	283	456.7			TUIGAMPX	10/02/2025

			Avg. 4.6019		38	284	460	50.33	A	TUIGAMPX	10/02/2025
=====											
92M 0108616013	0	1	4.6262	2.8356	38.7	284	462.6			TUIGAMPX	10/02/2025
		2	4.6272	2.8245	39.0	282	462.7			TUIGAMPX	10/02/2025
		3	4.5903	2.8233	38.5	282	459.0			TUIGAMPX	10/02/2025

			Avg. 4.6146		39	283	461	50.33	A	TUIGAMPX	10/02/2025
=====											
92M 0108616014	0	1	4.5502	2.8223	38.0	282	455.0			TUIGAMPX	10/02/2025
		2	4.5942	2.8260	38.5	283	459.4			TUIGAMPX	10/02/2025
		3	4.5338	2.8135	37.9	281	453.4			TUIGAMPX	10/02/2025

			Avg. 4.5594		38	282	456	50.33	A	TUIGAMPX	10/02/2025
=====											
92M 0108616015	0	1	4.5719	2.8076	38.6	281	457.2			TUIGAMPX	10/02/2025
		2	4.5825	2.8085	38.7	281	458.2			TUIGAMPX	10/02/2025
		3	4.5727	2.8156	38.4	282	457.3			TUIGAMPX	10/02/2025

			Avg. 4.5757		39	281	458	50.33	A	TUIGAMPX	10/02/2025
=====											
92M 0108616016	0	1	4.7277	2.8198	40.4	282	472.8			TUIGAMPX	10/02/2025
		2	4.7480	2.8263	40.5	283	474.8			TUIGAMPX	10/02/2025
		3	4.7287	2.8154	40.5	282	472.9			TUIGAMPX	10/02/2025

			Avg. 4.7348		40	282	474	50.33	A	TUIGAMPX	10/02/2025
=====											

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

Lot Averages Report for Lot: **92M0108616**

10/27/2025

Page 3 of 3

Spool	Retest:	RC	AWT	PAW	Stamp
92M0108616001	0	39	279	458	A
92M0108616002	0	38	283	457	A
92M0108616003	0	38	285	463	A
92M0108616004	0	39	282	459	A
92M0108616005	0	39	282	461	A
92M0108616006	0	38	284	460	A
92M0108616007	0	39	282	461	A
92M0108616008	0	38	282	458	A
92M0108616009	0	38	282	458	A
92M0108616010	0	38	283	458	A
92M0108616011	0	38	282	458	A
92M0108616012	0	38	284	460	A
92M0108616013	0	39	283	461	A
92M0108616014	0	38	282	456	A
92M0108616015	0	39	281	458	A
92M0108616016	0	40	282	474	A
Averages:		38.6	282.3	459.0	



Product Certification

Page 1 of 5

COA Number:	SEG-COA-28605	Prod Order:	DOM:	Description:
Version:	1	31W0028125	12-Jun-2025	F3C433(FBL2)-50"
Issue Date:	27-Jun-2025	Weave Pattern:	5 Harness Satin	

Specifications

F3C433(FBL2) - GREIGE - 50inch - HRMS 111546 Rev. 5 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex
F3C433(FBL2) - GREIGE - 50inch - MSRR4040 Issue 4 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex
F3C433(FBL2) - GREIGE - 50inch - MSRR9992 Issue 12 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

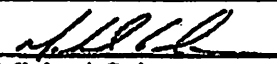
Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result:
31G0000373100	31W0028125	Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000373101		Width	49.5 - 50.5 in	50.3
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000373102		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000373103		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000373104		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7

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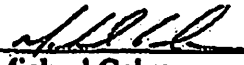
Product Certification

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31G0000373104	31W0028125	Dried FAW Avg	278 - 292 g/m ²	283
31G0000373105		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000373106		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373107		Width	49.5 - 50.5 in	50.3
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000373108		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000373109		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000373110		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373111		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000373112		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6

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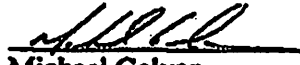
Product Certification

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31G0000373112	31W0028125	Dried FAW Avg	278 - 292 g/m ²	281
31G0000373113		Width	49.5 - 50.5 in	50.4
		Ends per inch	17.0 - 18.5 Yams Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yams Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373114		Width	49.5 - 50.5 in	50.3
		Ends per inch	17.0 - 18.5 Yams Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yams Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000373115		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yams Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yams Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373116		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yams Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yams Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000373117		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yams Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yams Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373118		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yams Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yams Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000373119		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yams Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yams Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000373120		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yams Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yams Per Inch	17.7

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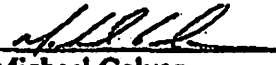
Product Certification

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31G0000373120	31W0028125	Dried FAW Avg	278 - 292 g/m ²	283
31G0000373121		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000373122		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373123		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373124		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000373125		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373126		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000373127		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373128		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7

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31G0000373128	31W0028125	Dried FAW Avg	278 - 292 g/m ²	282
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Certification Exceptions/Additional Information

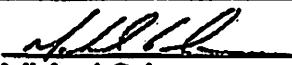
QMS07045, REV. 1, SEC. 5E

WARP: 250120210

FILL: 250120210

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Michael Galvan
Certification Technician

TEIJIN

TEIJIN CARBON AMERICA, INC.

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (865) 384-8408
Fax (865) 384-8409

Certificate of Analysis

QAP 004.F3 Rev M

Customer:	Hexcel Reinforcement Corp (TX)	Contact:	Patrick Spickermann; Amy Shehane; Jerry Tschoepe; Greg Early	Email/Fax:	Patrick.Spickermann@hexcel.com; amy.shehane@hexcel.com; jerry.tschoepe@; Greg.Early@
CustomerPO#:	31PO0014115	Order:	SH001658	Ship Date:	8/4/2025
Customer P/N:	n/a	Product:	Tenax™-E HTA40 E13 3K 260tax		
Part Number:	G305303	Quantity:	3144.000 kg	Certifies To:	MS-X-CTC-105 Rev B; MSRR9317/11 Issue 1
Lot Number:	250120210				

We are pleased to advise that the product(s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	Average	cv%
Tensile Strength, MPA (ksi):	4145.8 (601)	4.2
Tensile Modulus, GPA (msi):	238.4 (34.6)	0.8
Elongation, %:	1.64	4.0
Linear Density (without size), g/m:	0.1894	1.0
Size, %:	1.27	7.3
Fiber Density, g/cc:	1.77	
Filament Diameter, µm:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	January 2025	
Manufacturing Location:	Obertruch, Hainsberg, Germany	


Quality Assurance

8/4/2025

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If goods are not received in a reasonable time or if there are any questions, please call 1-800-252-3001 (or (865) 384-8408).
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood, TN 37854.



Product Certification

Page 1 of 2

COA Number:	SEG-COA-28997	Prod Order:	DOM:	Description:
Version:	1	31W0028125	12-Jun-2025	F3C433(FBL2)-50"
Issue Date:	19-Sep-2025	Weave Pattern:	5 Harness Satin	

Specifications

F3C433(FBL2) - GREIGE - 50inch - HRMS 111546 Rev. 6 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50inch - MSRR9992 Issue 12-Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

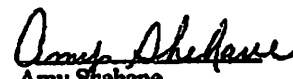
Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result:
31G0000373135	31W0028125	Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000373138		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373139		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000373140		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000373141		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284

This is to certify that material shipped against this order meets the specification(s) listed above.

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1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Amy Shehane
Administrative Assistant



Product Certification

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Certification Exceptions/Additional Information

QMS07045 REV. 1 SEC. 5E

WARP: 250120210

FILL: 250120210

This is to certify that material shipped against this order meets the specification(s) listed above.

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Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Amy Shehane
Administrative Assistant



TEIJIN CARBON AMERICA, INC.

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (865) 354-8408
Fax (865) 354-8409

Certificate of Analysis

QAP 004F3 Rev M

Customer:	Hexcel Reinforcement Corp (TX)	Contact:	Patrick Spieckermann; Amy Shehane; Jerry Tschoepe; Greg Early	Email/Fax:	Patrick.Spieckermann@hexcel.com; amy.shehane@hexcel.com; jerry.tschoepe@; Greg.Early@
Customer PO#:	31PO0014283	Order:	SH001734	Ship Date:	7/1/2025
Customer P/N:	n/a	Product:	Tenax™-E HTA40 E13 3K 200tex		
Part Number:	G305303	Quantity:	10000.086 lb	Certifies To:	MS-X-CTC-105 Rev B; MSRR9317/11 Issue 1
Lot Number:	250120210				

We are pleased to advise that the product (s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	<u>Average</u>	<u>cv%</u>
Tensile Strength, MPA (ksi):	4145.8 (601)	4.2
Tensile Modulus, GPA (msi):	238.4 (34.6)	0.8
Elongation, %:	1.64	4.0
Linear Density (without size), g/m:	0.1894	1.0
Size, %:	1.27	7.3
Fiber Density, g/cc:	1.77	
Filament Diameter, µm:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	January 2025	
Manufacturing Location:	Oberbruch, Heinsberg, Germany	


Quality Assurance 6/30/2025

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If goods are not received in a reasonable time or if there are any questions, please call 1-800-282-3001 (or (865) 354-8408).
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood, TN 37854.

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0108616	92M0108616001	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
6	50	2	0.7	CURED RESIN
11	50	3	1.0	CURED RESIN
41	50	0	0.0	FUZZBALL
44	50	0	0.0	FUZZBALL
53	50	0	0.0	FUZZBALL
59	50	0	0.0	FUZZBALL
70	50	0	0.0	FUZZBALL
79	50	0	0.0	FUZZBALL
88	50	0	0.0	FUZZBALL
114	50	0	0.0	FUZZBALL
163	50	0	0.0	FUZZBALL
197	50	4	1.4	WRINKLES
208	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
210 206	209 205	A	0.357%	10/2/2025



OCT 04 2025



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Inspection Stamp	Date
	10/2/2025

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0108616	92M0108616002	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
47	50	1	0.3	MISPICK
120	50	24	8.3	CURED RESIN
124	50	1	0.3	CURED RESIN

Final Footage	Net Feet	Credit Method	Defect %	Date
150	147	A	1.444%	10/2/2025

Inspection Stamp	Date
	10/2/2025

ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M0108616003	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type	Remarks
17	50	1	0.3	MISPICK	
131	50	84	29.2	SNAG	

Final Footage 200	Net Feet 192	Credit Method A	Defect % 3.542%	Date 10/2/2025
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Inspection Stamp 	Date 10/2/2025
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ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0108616	92M0108616004	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
86	50	12	4.2	SNAG
203	50	1	0.3	MISPICK

Final Footage	Net Feet	Credit Method	Defect %	Date
250	248	A	0.433%	10/2/2025

Inspection Stamp	Date
	10/2/2025

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0108616	92M01086005	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type	Remarks
11	50	0	0.0	FUZZBALL	
18	50	0	0.0	FUZZBALL	
88	50	1	0.3	SLUB	
200	50	24	8.3	CURED RESIN	
205	50	1	0.3	CURED RESIN	
211	50	1	0.3	CURED RESIN	

Final Footage	Net Feet	Credit Method	Defect %	Date
225	222	A	1.000%	10/2/2025

Inspection Stamp	Date
	10/2/2025

ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M010861006	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
10	50	0	0.0	FUZZBALL
40	50	12	4.2	CREASE
45	50	0	0.0	FUZZBALL
59	50	0	0.0	FUZZBALL
94	50	24	8.3	SPLICE
101	50	12	4.2	SPLICE
104	50	0	0.0	FUZZBALL
105	50	24	8.3	CURED RESIN
121	50	1	0.3	CURED RESIN
131	50	1	0.3	CURED RESIN
211	50	48	16.7	SNAG

Final Footage 224	Net Feet 213	Credit Method A	Defect % 4.539%	Date 10/2/2025
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Inspection Stamp 	Date 10/2/2025
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ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M0108616007	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
100	50	0	0.0	FUZZBALL
135	50	0	0.0	FUZZBALL
144	50	2	0.7	RESIN LEAN
153	50	0	0.0	FUZZBALL
173	50	0	0.0	FUZZBALL

Final Footage 225	Net Feet 224	Credit Method A	Defect % 0.074%	Date 10/2/2025
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Inspection Stamp 	Date 10/2/2025
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ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0108616	92M0108616008	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
33	50	1	0.3	SLUB
53	50	0	0.0	FUZZBALL
134	50	0	0.0	FUZZBALL
211	50	1	0.3	CURED RESIN
216	50	1	0.3	CURED RESIN

Final Footage	Net Feet	Credit Method	Defect %	Date
220	219	A	0.114%	10/5/2025

Inspection Stamp	Date
	10/5/2025

ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M0108616009	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type	Remarks
88	50	0	0.0	FUZZBALL	
130	50	0	0.0	FUZZBALL	

Final Footage 203	Net Feet 203	Credit Method A	Defect % 0.000%	Date 10/5/2025
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Inspection Stamp 	Date 10/5/2025
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ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M0108616010	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
211	50	12	4.2	RESIN LEAN

Final Footage 236	Net Feet 235	Credit Method A	Defect % 0.424%	Date 10/2/2025
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Inspection Stamp 	Date 10/2/2025
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ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0108616	92M0108616011	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
104	50	1	0.3	MISPICK
135	50	0	0.0	FUZZBALL
166	50	0	0.0	FUZZBALL
198	50	24	8.3	CURED RESIN
205	50	1	0.3	CURED RESIN
209	50	1	0.3	CURED RESIN

Final Footage	Net Feet	Credit Method	Defect %	Date
225	222	A	1.000%	10/2/2025

Inspection Stamp	Date
	10/2/2025

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0108616	92M0108616012	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
70	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
225	225	A	0.000%	10/2/2025

Inspection Stamp	Date
	10/2/2025

ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M0108616013	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type	Remarks
8	50	0	0.0	FUZZBALL	
113	50	3	1.0	FLOAT	
198	50	0	0.0	FUZZBALL	

Final Footage 225	Net Feet 224	Credit Method A	Defect % 0.111%	Date 10/4/2025
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Inspection Stamp 	Date 10/4/2025
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ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M01056014	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
39	50	0	0.0	FUZZBALL
60	50	0	0.0	FUZZBALL

Final Footage 224	Net Feet 224	Credit Method A	Defect % 0.000%	Date 10/2/2025
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Inspection Stamp 	Date 10/2/2025
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ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M0108616015	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
10	50	0	0.0	FUZZBALL
45	50	1	0.3	FLOAT
67	50	0	0.0	FUZZBALL
109	50	1	0.3	SLUB

Final Footage 225	Net Feet 224	Credit Method A	Defect % 0.074%	Date 10/2/2025
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Inspection Stamp 	Date 10/2/2025
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ROLL CONDITION LOG

Batch Number 92M0108616	Serial Number 92M0108616016	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage 225 191	Net Feet 225 191	Credit Method A	Defect % 0.000%	Date 10/2/2025
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OCT 04 2025



OCT 04 2025

Inspection Stamp 	Date 10/2/2025
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